

LESSON 7-7 • SESSION 2 • PREREQUISITE PRACTICE B • SKILL 1 OF 2

Use Equivalent Ratios for Scale

Builds on Lesson 3-3 • 6.RP.A.3a

TODAY'S LESSONOn a plan, 1 unit = 6 ft. So 4 units is $4 \times 6 = 24$ ft.**YOU NEED TO**Change grid units to feet with a scale, like 1 unit = 25 ft. **You will use it in Practice Form A, problems 1 and 4.****STEPS TO FOLLOW***use these same steps on every problem*

- 1 Read** the scale: 1 unit = how many feet.
- 2** Units to feet: **multiply**. Feet to units: divide.
- 3 Label** the answer with feet or units.

WATCH ME*the steps in action***Scale: 1 unit = 25 ft. A side is 6 units long. How long is it in feet?****Step 1** 1 unit = 25 ft.**Step 2** Units to feet: $6 \times 25 = 150$.**Step 3** The side is 150 ft.**Answer 150 ft****TRY IT WITH HELP***fill in each step*

- 1.** Scale: 1 unit = 10 ft. A side is 7 units. How many feet?

Step 1 1 unit = _____ ft.**Step 2** Units to feet: $7 \times 10 =$ _____.**Step 3** The side is _____ ft.

- 2.** Scale: 1 unit = 5 ft. A wall is 40 ft. How many units?

Step 1 1 unit = _____ ft.**Step 2** Feet to units: $40 \div 5 =$ _____.**Step 3** The wall is _____ units.**YOUR TURN***use the steps on your own*

- 3.** Change each measure. Write ft or units.

a. 1 unit = 3 ft; 12 units _____**b.** 1 unit = 4 ft; 9 units _____**c.** 1 unit = 15 ft; 8 units _____**d.** 1 unit = 10 ft; 80 ft _____**e.** 1 unit = 25 ft; 175 ft _____**f.** 1 unit = 6 ft; 42 ft _____

Perimeter and Area on Grids

Builds on Lesson 7-7 Session 1 • 6.G.A.3

TODAY'S LESSONThe parking pad side from $(-7, 5)$ to $(5, 5)$ is 12 units long.**YOU NEED TO**Find a grid rectangle's length, width, perimeter and area in units. **You will use it in Practice Form A, problems 2, 3 and 6.****STEPS TO FOLLOW***use these same steps on every problem*

- 1** Find the **length**: the distance across the top side.
- 2** Find the **width**: the distance down the right side.
- 3** Find **perimeter** (add four sides) and area (length $\times$ width).

WATCH ME*the steps in action***A rectangle has vertices $(-3, 4)$, $(5, 4)$, $(5, -2)$ and $(-3, -2)$. Find its perimeter and area.****Step 1** Length: $(-3, 4)$ to $(5, 4)$ is $3 + 5 = 8$ units.**Step 2** Width: $(5, 4)$ to $(5, -2)$ is $4 + 2 = 6$ units.**Step 3** $P = 8 + 6 + 8 + 6 = 28$ units. $A = 8 \times 6 = 48$ square units.**Answer** **$P = 28$ units, $A = 48$ square units****TRY IT WITH HELP***fill in each step***4.** Vertices: $(-2, 3)$, $(4, 3)$, $(4, -1)$ and $(-2, -1)$.**Step 1** Length: $2 + 4 =$ _____ units.**Step 2** Width: $3 + 1 =$ _____ units.**Step 3** $P =$ _____ units. $A =$ _____ square units.**5.** Vertices: $(-4, 5)$, $(3, 5)$, $(3, -3)$ and $(-4, -3)$.**Step 1** Length: $4 + 3 =$ _____ units.**Step 2** Width: _____ $+ 3 = 8$ units.**Step 3** $P =$ _____ units. $A =$ _____ square units.**YOUR TURN***use the steps on your own***6.** A rectangle has vertices $(-6, 4)$, $(4, 4)$, $(4, -2)$ and $(-6, -2)$.**a.** length _____ **b.** width _____ **c.** perimeter _____ **d.** area _____**READY CHECK***a problem like the one in today's lesson***7.** Corners: $(-5, 3)$, $(7, 3)$, $(7, -2)$ and $(-5, -2)$. What is the area?**A** 17 square units **B** 34 square units **C** 60 square units **D** 10 square units**CHECK YOURSELF**
☐ I can do this on my own
 ☐ I need a little help
 ☐ I have a question about #____